

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	20% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024-AI&ML	Date:	17.08.2026

Code of Conduct

I D.Sravanthi(192425315) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D.Sravanthi

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A language processing system must represent sentence structure for grammar checking and syntactic analysis. The developers are deciding between Context-Free Grammar (CFG) trees and dependency parsing structures. Analyze how these two approaches differ in representing sentence structure and justify which is more effective for capturing relationships between words.

Answer:

CFG Trees vs. Dependency Parsing

Context-Free Grammar (CFG) trees represent a sentence using hierarchical constituents such as noun phrases (NP), verb phrases (VP), and prepositional phrases (PP). They show how words combine to form larger grammatical units.

Dependency parsing, on the other hand, represents relationships directly between words. Each word is connected to its syntactic head, showing relations such as subject, object, modifier, and complement.

Aspect	CFG Trees	Dependency Parsing
Main focus	Phrase structure	Word-to-word relationships
Representation	Hierarchical tree	Directed dependency tree
Example	NP → Det + N	<i>man</i> → object of <i>saw</i>
Long-distance relations	Less direct	Easier to represent
Useful for	Grammar and constituency analysis	Semantic and syntactic relationships

Justification: Dependency parsing is generally more effective for capturing relationships between words because it directly identifies which word depends on another. It is particularly useful for information extraction, question answering, and semantic analysis.

Conclusion: CFG is better for understanding phrase structure, while dependency parsing is more effective for representing direct grammatical relationships between words.

2. A real-time application needs to parse user input efficiently, even when sentences are incomplete or ambiguous. The system currently uses top-down parsing, but an alternative is Earley parsing. Analyze the differences between these parsing techniques and reason which is more suitable for such dynamic input conditions.

Answer:

Top-Down Parsing vs. Earley Parsing

Top-down parsing starts from the start symbol of the grammar and attempts to generate the input sentence by predicting grammar rules. It can be simple and efficient for well-formed and predictable sentences, but it may struggle with ambiguity and incomplete input.

Earley parsing is a chart-parsing technique that combines prediction, scanning, and completion. It can handle arbitrary context-free grammars and is capable of dealing with ambiguous and incomplete sentences more flexibly.

Aspect	Top-Down Parsing	Earley Parsing
Starting point	Grammar start symbol	Grammar + input positions
Ambiguity handling	Limited	Good
Incomplete input	Less flexible	More suitable
Grammar support	Often requires suitable grammar	General CFG
Memory	Can repeatedly explore alternatives	Uses chart to avoid repetition
Dynamic input	Less suitable	More suitable

Justification: For real-time applications where users may enter incomplete or ambiguous sentences, **Earley parsing is more suitable**. Its chart structure stores intermediate results and prevents unnecessary repeated parsing.

Conclusion: Top-down parsing is useful for simple and predictable inputs, whereas Earley parsing is preferable for dynamic, incomplete, and ambiguous user input.

3. An NLP model must handle ambiguity in sentences such as “She saw the man with a telescope.” The designers are considering CFG, PCFG, and neural parsing techniques.

Analyze how these approaches differ in handling ambiguity and reason which method is most effective for real-world applications.

Answer:

CFG vs. PCFG vs. Neural Parsing

Consider the sentence:

“She saw the man with a telescope.”

This sentence has two possible interpretations:

1. She used a telescope to see the man.
2. The man had a telescope.

A **CFG** can represent both possible parse trees, but it normally cannot determine which interpretation is more likely.

A **Probabilistic CFG (PCFG)** assigns probabilities to grammar rules. Therefore, it can rank the possible interpretations and select the more probable parse.

Neural parsing uses machine-learning models to learn syntactic patterns and contextual information from large datasets. Modern neural parsers can use surrounding words and contextual representations to resolve many types of ambiguity.

Aspect	CFG	PCFG	Neural Parsing
Handles ambiguity	Represents alternatives	Ranks alternatives	Uses learned context
Probability	No	Yes	Learned scores
Context understanding	Low	Moderate	High
Training data	Usually not required	Requires probabilities/data	Requires large datasets
Real-world performance	Limited	Better	Generally strongest

Justification: Neural parsing is generally the most effective for real-world applications because it can learn complex contextual relationships and handle ambiguity more effectively than traditional CFG and PCFG approaches.

Conclusion: CFG provides the basic grammatical structure, PCFG improves ambiguity resolution through probabilities, while **neural parsing provides the best overall performance for modern NLP applications.**

4. A grammar system needs to correctly handle subject–verb agreement and verb argument structures. The developers are comparing feature structures and subcategorization frames. Analyze how these approaches differ and justify which provides better support for enforcing grammatical correctness.

Answer:

Feature Structures vs. Subcategorization Frames

Feature structures represent grammatical properties such as number, person, tense, gender, and case. They can enforce constraints such as subject–verb agreement. For example:

- **She runs** → singular subject + singular verb
- **They run** → plural subject + plural verb
- **She run** → agreement violation

Subcategorization frames describe what types of complements a verb requires.

For example:

- **eat** → requires a noun phrase: *She eats food.*
- **give** → can take a subject, object, and indirect object: *She gives him a book.*
- **depend** → normally requires a prepositional complement: *It depends on you.*

Aspect	Feature Structures	Subcategorization Frames
Main purpose	Grammatical features	Verb argument requirements
Subject–verb agreement	Strong support	Limited
Verb complements	Can represent	Strong support
Number/person	Directly represented	Usually not the focus
Grammatical constraints	Broad	Verb-specific

Justification: Feature structures provide better overall support for enforcing grammatical correctness because they can explicitly represent and unify features such as number and person. Subcategorization frames are particularly useful for checking whether a verb has the correct arguments and complements.

Conclusion: Feature structures are better for grammatical agreement, while subcategorization frames are better for validating verb argument structures. A combination of both provides the strongest grammar system.

5. A large-scale NLP system must perform fast and accurate dependency parsing on big datasets. The choice is between transition-based parsing and graph-based parsing.

Analyze the differences between these methods in terms of decision-making and performance, and justify which is more suitable for large-scale applications.

Answer:

Transition-Based vs. Graph-Based Dependency Parsing

Transition-based parsing constructs a dependency tree step by step using a sequence of actions such as SHIFT, REDUCE, and ARC. Each decision determines the next parsing operation.

Graph-based parsing considers possible dependency relationships between words and searches for the highest-scoring complete dependency tree.

Aspect	Transition-Based	Graph-Based
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Aspect	Transition-Based	Graph-Based
Decision-making	Local, step-by-step	Global optimization
Speed	Very fast	Generally slower
Accuracy	Can suffer from early errors	Usually more robust
Error propagation	Higher	Lower
Large-scale processing	Excellent	More computationally expensive
Long-distance dependencies	Can be challenging	Generally handles them well

Justification: For large-scale NLP systems, **transition-based parsing is generally more suitable when speed and scalability are the primary requirements.** It has fast inference and relatively low computational cost, making it appropriate for processing millions of sentences.

However, if maximum parsing accuracy is more important than speed, graph-based parsing can be preferred because it evaluates the dependency structure more globally.

Conclusion: Transition-based parsing is the better choice for **large-scale, real-time applications**, while graph-based parsing is preferable when higher global accuracy is required.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____